

FORENSIC INVESTIGATION REPORT

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1. Executive Summary

This report documents the forensic investigation and successful recovery of embedded data from a corrupted disk image, "**carveit**". As a junior digital forensics analyst, the objective was to identify, extract, and validate lost or hidden files using automated carving techniques. The investigation resulted in the recovery of **18 unique files**, including images, documents, and archives, all of which were verified for integrity using MD5 cryptographic hashing.

2. Investigation Methodology

The investigation followed a structured forensic workflow to ensure data integrity and accuracy:

- **Initial Triage:** Performed binary analysis using hex examination tools to identify potential file signatures and data structures within the 16MB image.
- **Signature Identification:** Identified key forensic markers, including headers and footers for JPEG, PNG, PDF, ZIP, and GIF file formats.
- **Automated Carving:** Developed a custom Python script, `Mwaura_carver.py`, to automate the scanning and extraction process.
- **Integrity Verification:** Generated MD5 hashes for every recovered file to establish a baseline for forensic evidence preservation.

3. Technical Implementation

The primary tool used was a custom-built Python script designed to perform bit-stream carving. The script identifies the **Magic Bytes** (headers) and corresponding **Footers** to reconstruct complete files from the unallocated space of the disk image.

Signature Reference Table

FILE TYPE	HEADER (HEX)	FOOTER (HEX)
JPEG	FF D8 FF	FF D9
PNG	89 50 4E 47 0D 0A 1A 0A	49 45 4E 44 AE 42 60 82
PDF	25 50 44 46	25 25 45 4F 46
ZIP	50 4B 03 04	50 4B 05 06

4. Recovered Files & Metadata

The automated scan successfully recovered 18 files. Below is a summary of the recovered evidence:

FILE NAME	TYPE	OFFSET (BYTES)	SIZE (BYTES)	INTEGRITY STATUS
recovered_001.jpg	JPEG	117,768	40,777	Verified
recovered_013.png	PNG	758,970	1,202,798	Verified
recovered_016.pdf	PDF	495,315	31,737	Verified
recovered_017.zip	ZIP	1,961,982	246,115	Verified

FILE NAME	TYPE	OFFSET (BYTES)	SIZE (BYTES)	INTEGRITY STATUS
recovered_018.gif	GIF	14,261,356	4,619	Verified

*Note: A full manifest of all 18 files and their MD5 hashes is provided in the accompanying *hashes.txt* file.

5. Challenges and Solutions

During the investigation, several challenges were encountered:

- **Data Fragmentation:** Some files appeared to be non-contiguous. The script was refined to search for the closest valid footer after each header to minimize corruption.
- **False Positives:** Initial scans identified partial signatures. Validation logic was added to ensure minimum file sizes and proper header-footer pairing.
- **Mock Data Padding:** The disk image contained significant ASCII padding (space characters), which initially obscured the data. Python's `rb` (read binary) mode was essential to bypass this and access the raw hex.

6. Conclusion and Recommendations

The forensic carving process was highly successful, yielding 18 fully intact files. The use of automated scripting allowed for a repeatable and verifiable recovery process, meeting the standards for digital forensic evidence. It is recommended that the recovered files be further analyzed for content and

that the disk image be kept in a read-only state to prevent any further data degradation.